

Software installation (Windows)

Summary

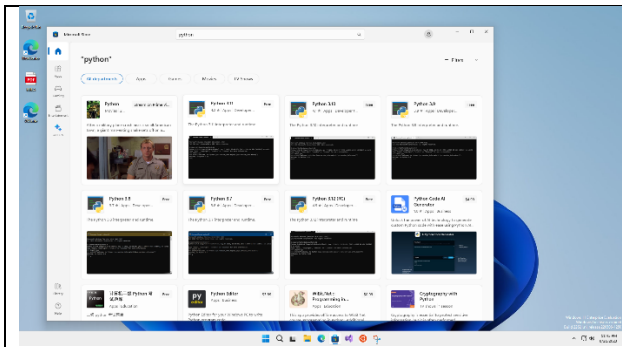
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This short document describes the fundamental steps to install the software used in the Computer Science course: the Python language interpreter, and the Visual Studio Code development environment.

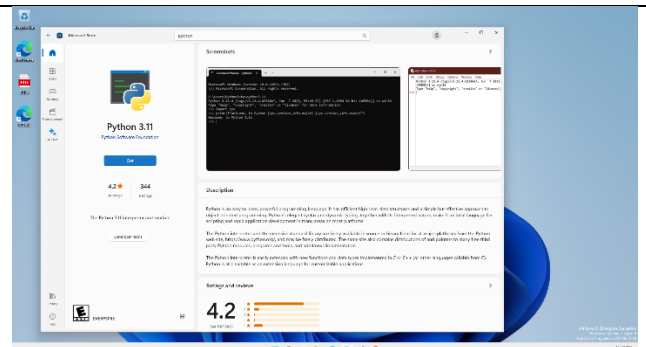
*The following instructions are for installation on a **Windows operating system**.*

Installing the Python Interpreter

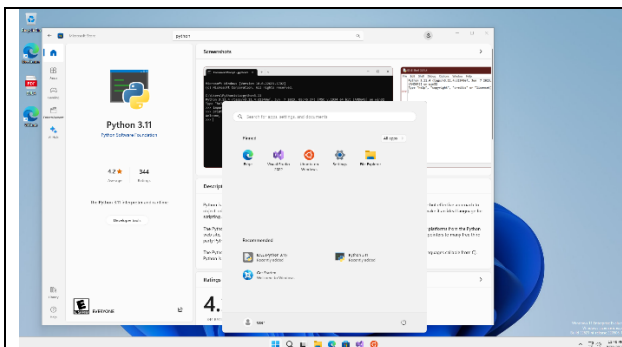
Recommended option: Install Python from the Microsoft Store



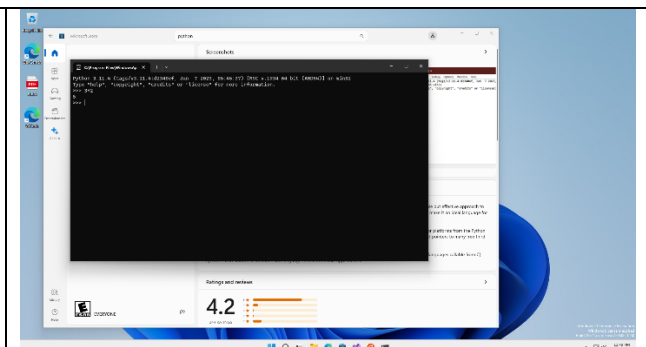
1. Open the “Microsoft Store” application and search for “Python” in the search box



2. Select Python **3.12** (from the Python Software Foundation), avoiding the 'fakes' that sometimes appear.
Select Get and wait for the installation.



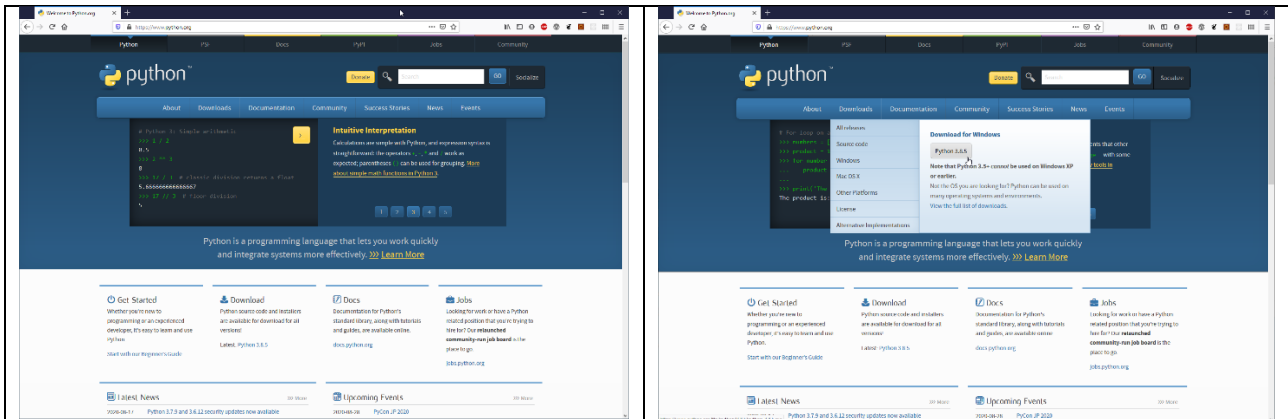
3. Once installed, the “Python **3.12**” icon will appear in the start menu



4. To verify correct installation, **open the Python window**.

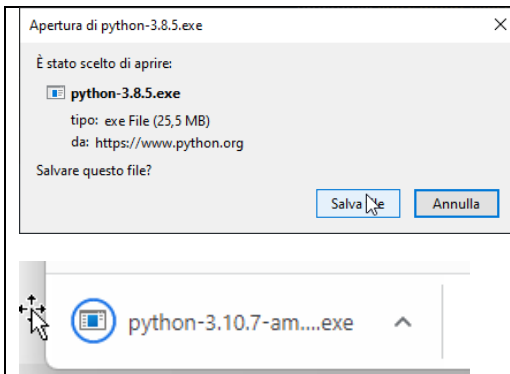
The first line will display the version number of the Python interpreter you just installed.
Write any expression (e.g. 3+2) and Python will print the result.
To finish, close the window or type Ctrl+Z and then Enter

Advanced option : Install Python from the Python.org website

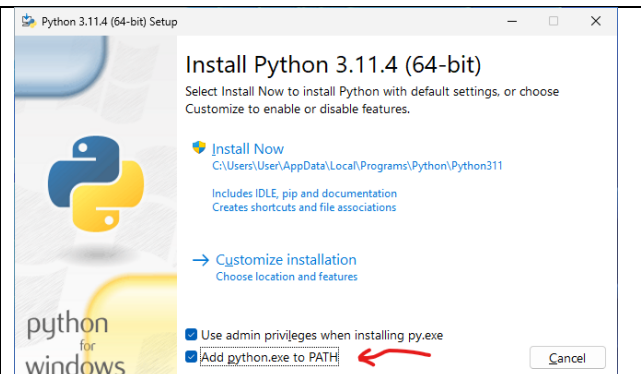


1. Navigate to <https://www.python.org>

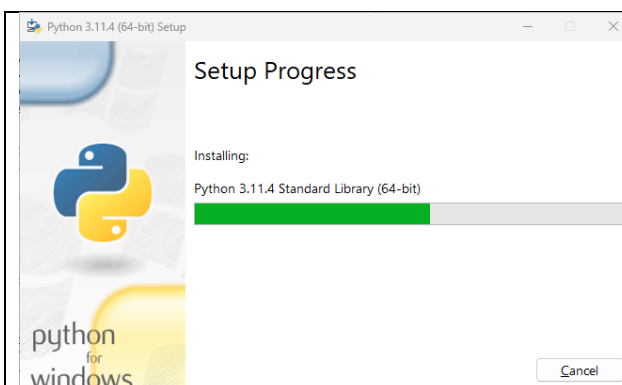
2. Select **Downloads** and then the version of Python appropriate for your operating system.
Note: It is recommended to install the latest version (currently 3.12.x). For Windows 7 PCs, you can only install version 3.8.



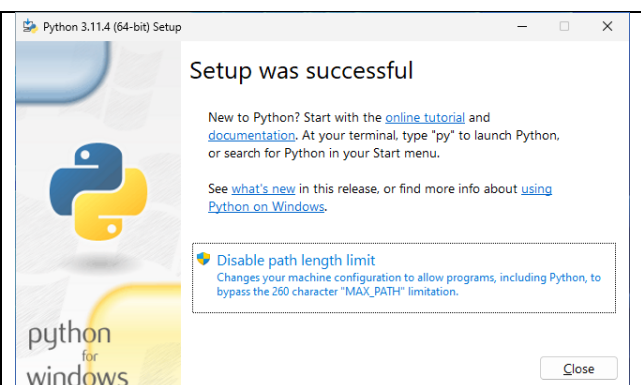
3. Download and save the installer



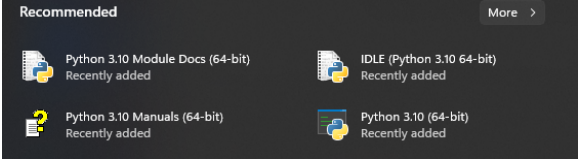
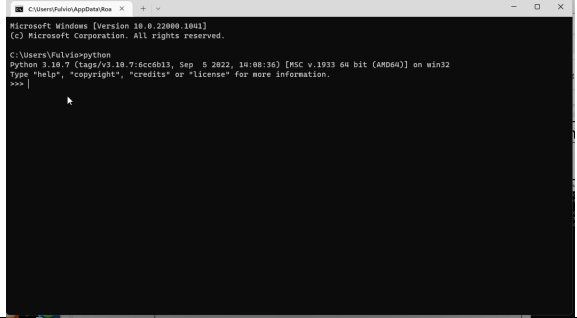
4. Start the installer
5. Check the “ Add python.exe to PATH” box
(makes it easier to launch Python from 'Command Prompt' windows)
6. Select “ Install Now ” to start the installation



7. Wait for the installation process to complete

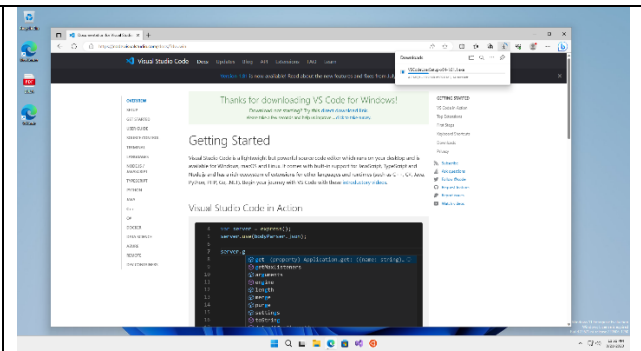
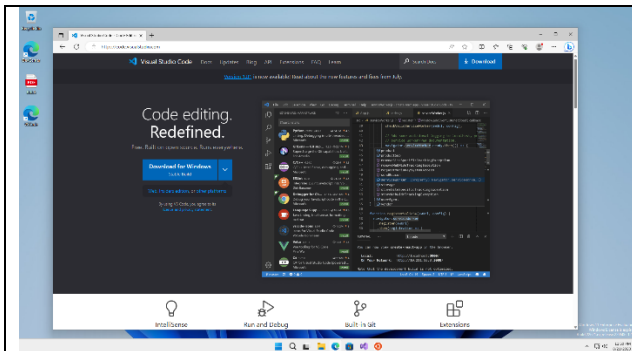


8. Select the “ Disable” option path length limit ”, if it is proposed.
9. Close the installer

	
10. Python related entries should have appeared in the Windows start menu	11. Open the “Command Prompt” from Windows applications 12. Inside the “Command Prompt” window, type the command python and verify that a startup message like the one shown above appears. This is the interactive Python interpreter. 13. To exit the interpreter, type Ctrl -Z and Enter

Installing Visual Studio Code

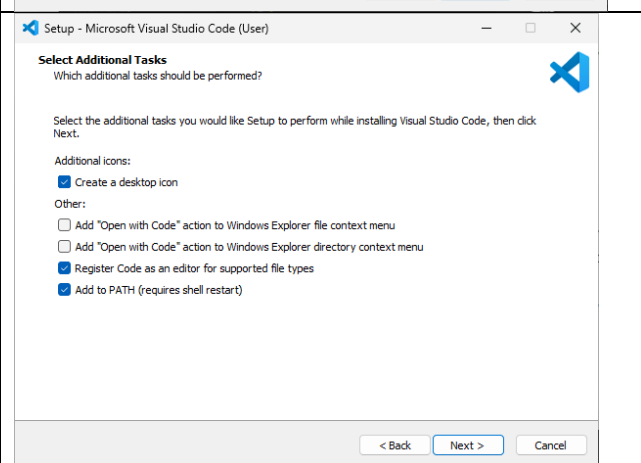
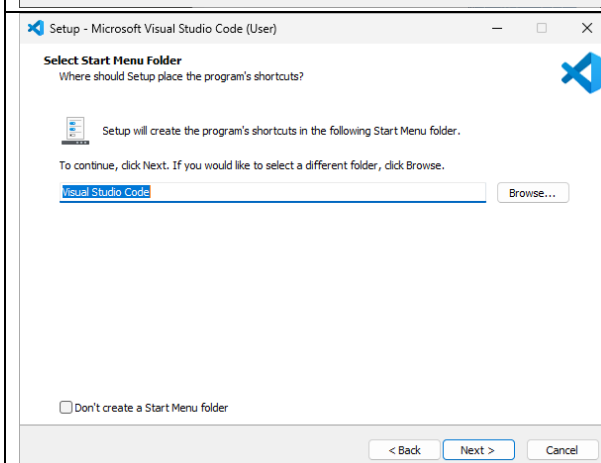
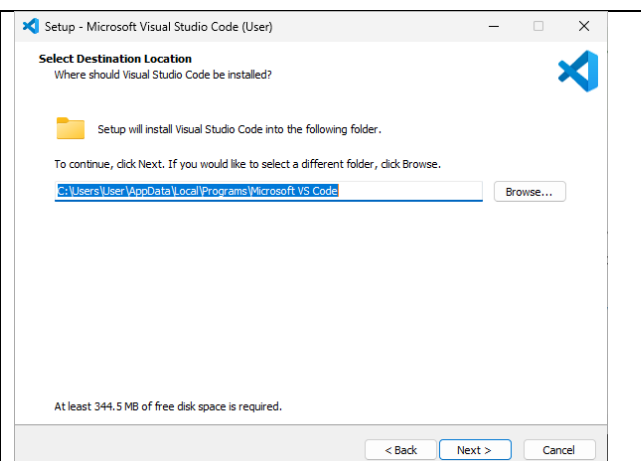
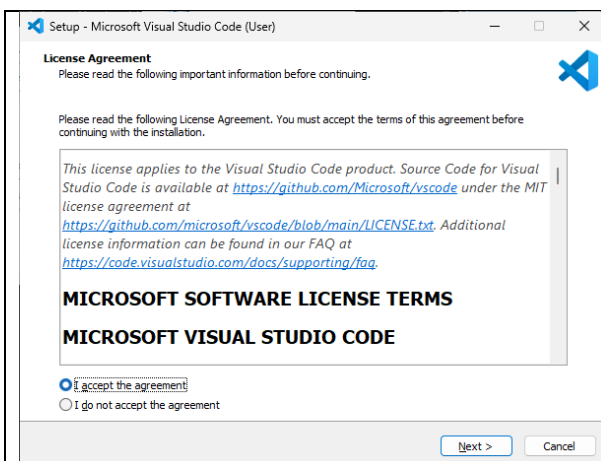
The environment in which we will write and run Python programs (IDE – Integrated Development Environment) is called Visual Studio Code.

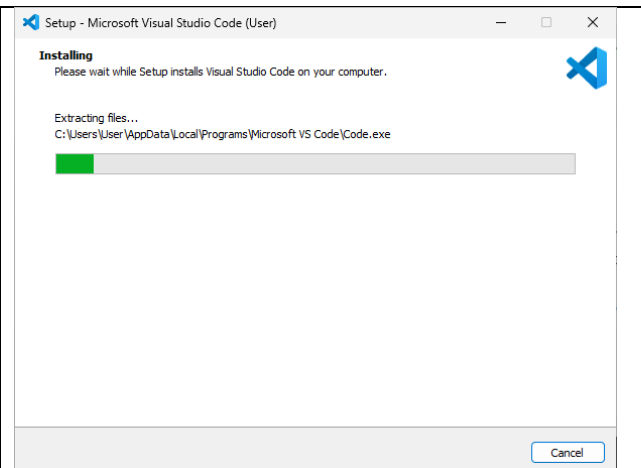
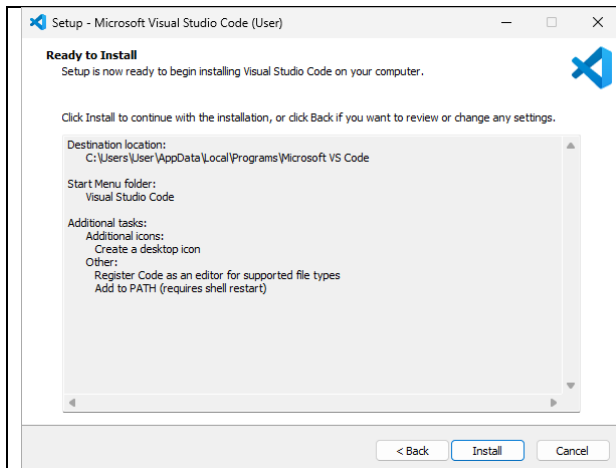


1. Navigate to <https://code.visualstudio.com/>

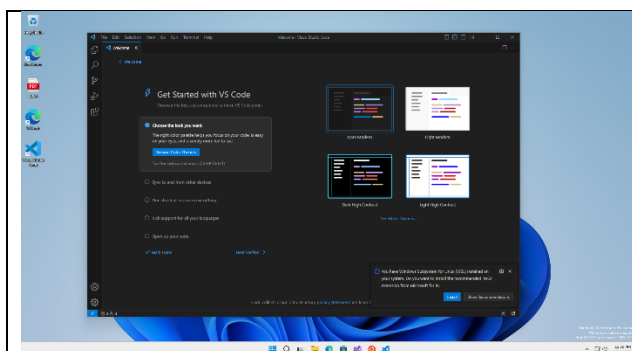
2. Download VS Code

3. Once the download is complete, launch the installer.

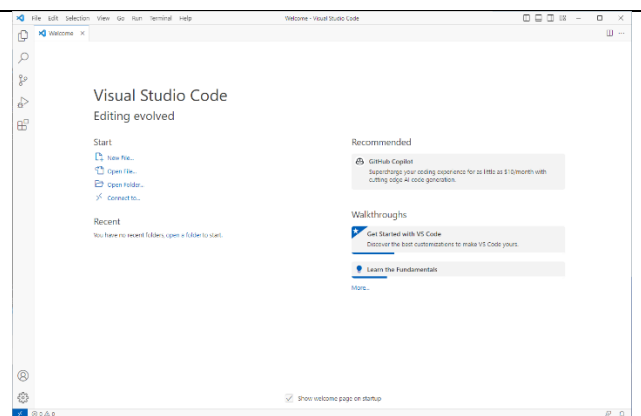




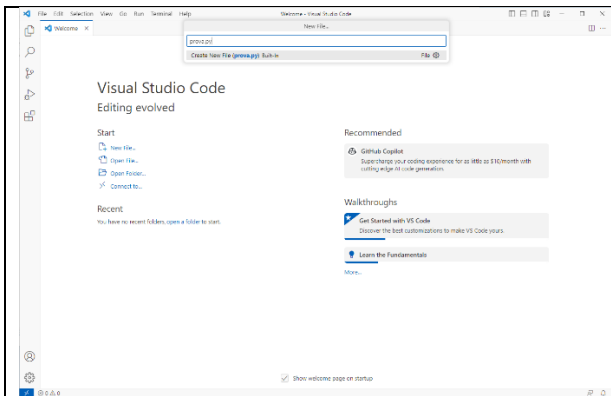
4. Follow the steps suggested by the installation program (all the values suggested by the wizard are fine)
Once finished, launch Visual Studio Code



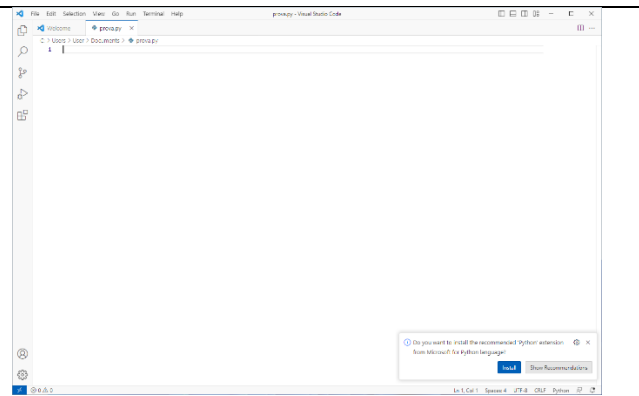
5. At first launch, you need to select some settings (such as light/dark theme)



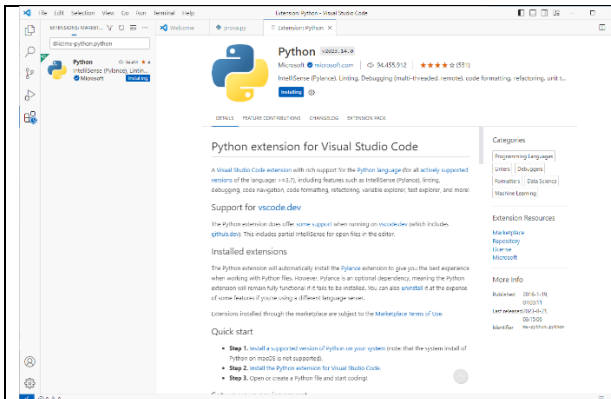
6. Once the initial settings are complete, you will find yourself on the main page (empty, because we have not yet opened any projects)



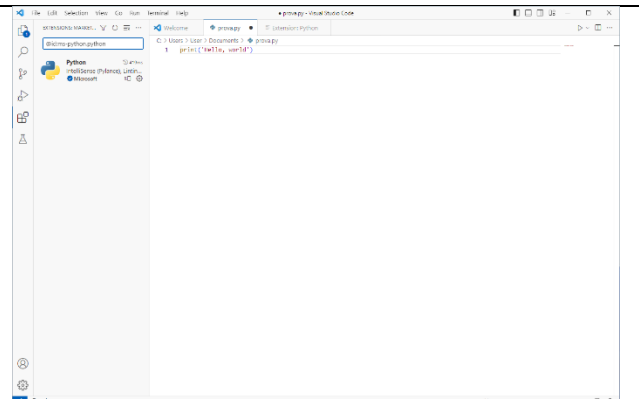
7. Select “New File” and enter a file name (for example prova.py). It is important that the file extension is **.py**. Save the file in a folder of your choice.



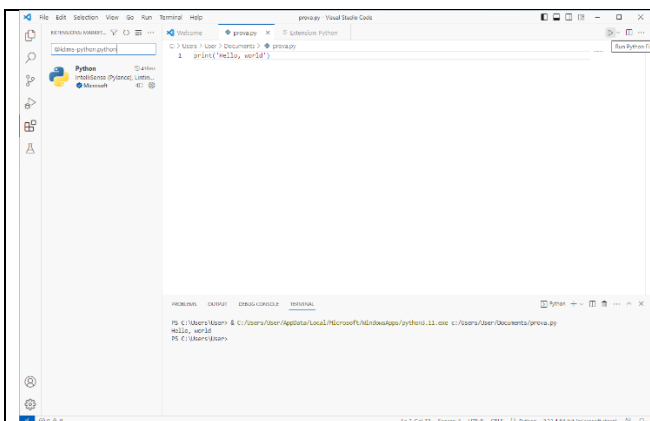
8. VS Code understands (from the .py extension) that we want to program in Python, and suggests (bottom right) that we install the Extensions for editing Python projects. Install the extension that is offered. Please note: this automation only happens the first time, otherwise you need to manually install the Python extension.



9. Proceed with the installation of the Python extension, and wait for the installation to finish.



10. Write a small Python program, for example: `print ( "Hello, world" )`



Run the program by clicking the triangle “Play” button at the top right of the screen. At the bottom, you will see the program output (`Hello, world`), or any error messages.

Python Interpreters Online

There are several websites where you can write and run Python programs directly from your browser, without installing programs on your PC. They are very useful tools both for doing tests "on the fly" and for sharing programs with teachers or other students.

But remember to save/export/download your work to your PC too, to avoid losing it.

- Replit.com (<https://replit.com/>) is a collaborative environment where you can write, test and share your code (in Python and other languages). Creating an account is recommended, also for the ease of sharing your work.
- Python Tutor (<https://pythontutor.com/>) is not a real programming environment, but it allows you to execute some lines of code, displaying in a graphical and interactive way the variables and values stored in the program. Very nice way to understand how variables work in Python.

Other Python IDEs

Visual Studio Code is not the only IDE suitable for programming in Python. There are many others, for example:

- **IDLE** (Python Integrated Development and Learning Environment), automatically installed along with Python, extremely simple and minimal .
- **PyCharm** is a very complete and feature-rich product by JetBrains . There are 3 editions:
 - PyCharm Professional (<https://www.jetbrains.com/pycharm/>), a paid version (but by registering as a university student you can get a free annual license), the most complete version
 - PyCharm Community (<https://www.jetbrains.com/pycharm/>), a reduced version but free for everyone (the missing functions are mainly related to scientific libraries and web development)
 - PyCharm for Education (<https://www.jetbrains.com/pycharm-edu/>), a free, simplified version, useful for beginners